Teacher-in-the-Loop LLM Exercise Authoring for Cambridge KET and PET

澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

Student: Zeming Chen Supervisor: Philip Lei Assessor: Benedito
Bachelor of Science in Computing (2026-2027)
Macao Polytechnic University 31 October 2026

PROBLEM

Supply falls short.

Learners practise weekly for two to three years, and the official past papers run out well before that.

Authoring by hand is slow.

A teacher must hold the part's structure, the CEFR band and defensible distractors at once; harder item types get skipped.

General AI tools fall short.

Fluent output, but no exam structure, no vocabulary boundary, and no structured place for the teacher's judgement.

SOLUTION AND METHOD

The AI drafts. Program code checks. The teacher decides.

The teacher picks the exam and part. Program code checks the shape and the handbook counts, and a reader answers cloze without the key. The teacher edits any field or rejects with a reason.

Measure the approval

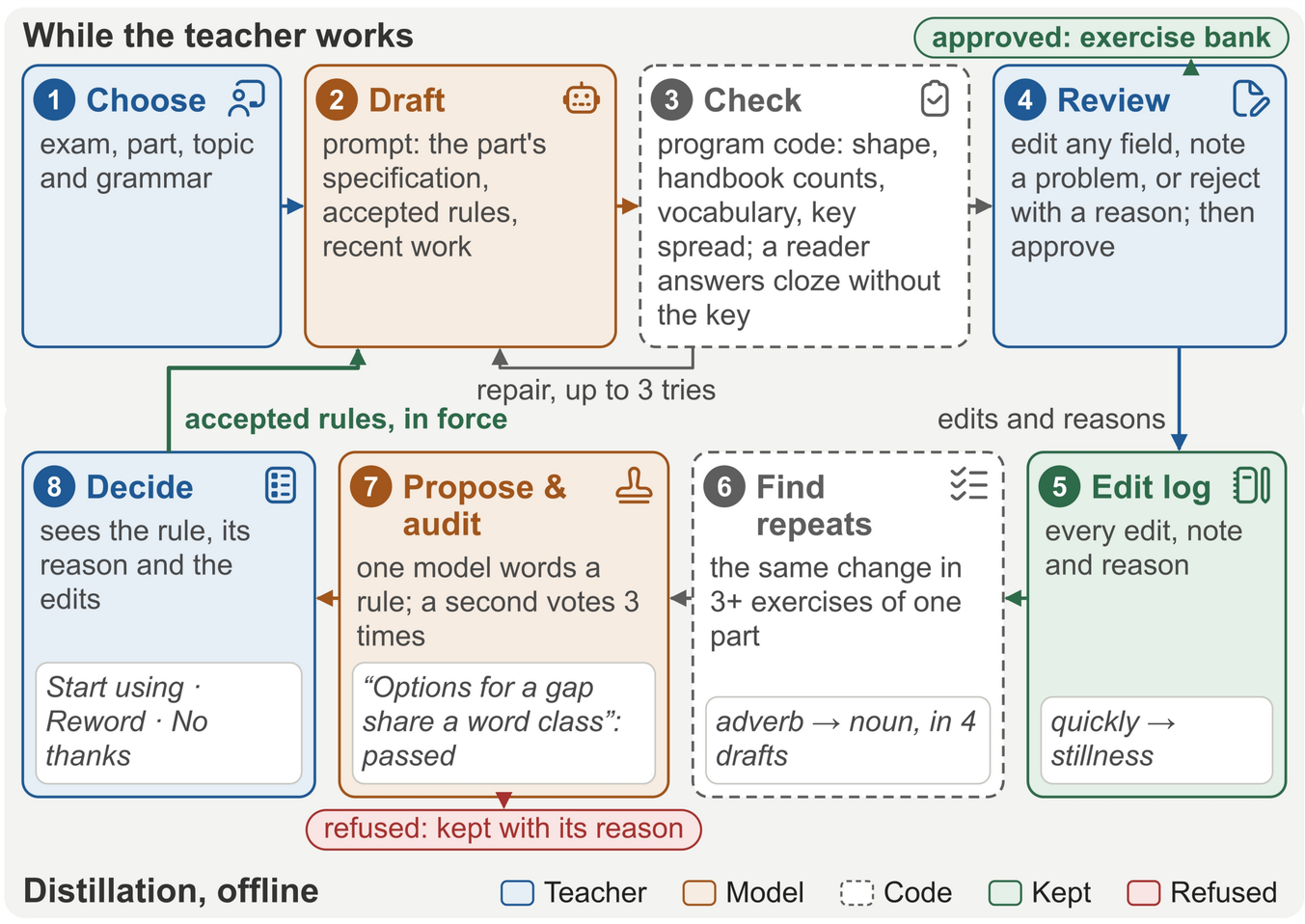
The untouched draft is frozen beside the edited one, so every correction is recorded at field level.

Distil offline, under the teacher's control

A change repeated in three or more exercises of one part becomes a proposed rule, worded by one model and voted on three times by a second; a refused rule is kept with its reason.

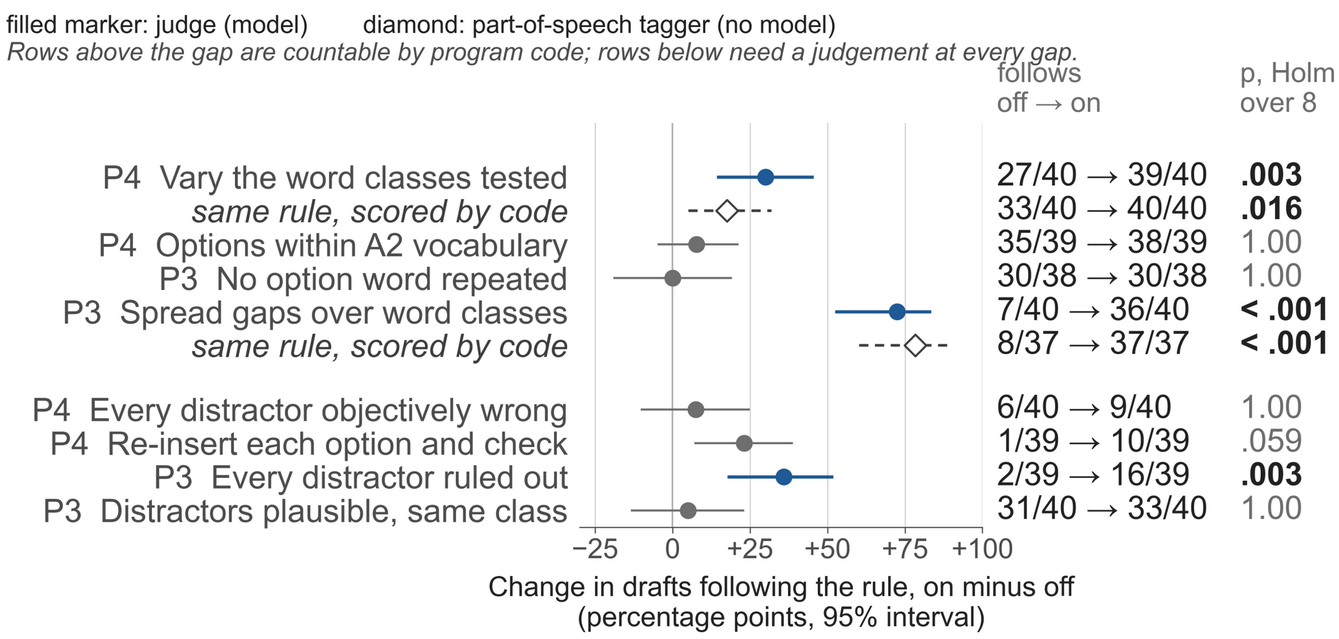
The teacher decides

Only rules the teacher accepts reach the next draft, and any rule can be stopped later.

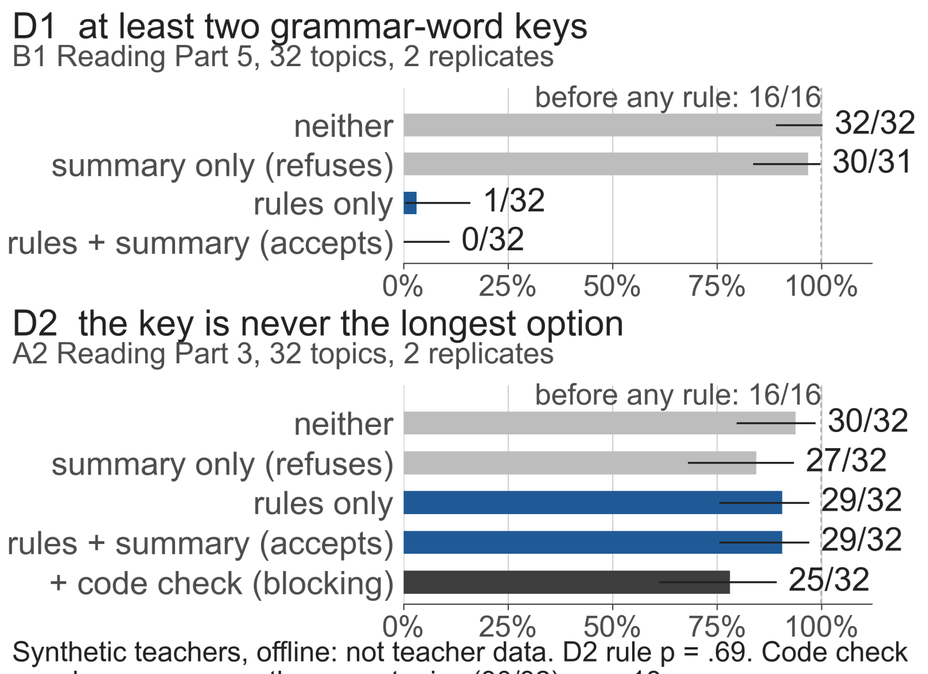


RESULTS

623 traces, **612** with a model call: **79.7%** needed no repair. First failed check: key/text **51**, reader **44**, handbook **19**, schema **11**. **54** regression scripts run the shipped code.



Scored by model judges, not teacher data.
Each rule measured with the teacher's four accepted rules on.
p: Holm over the eight judge tests; the tagger rows are unadjusted.



Synthetic teachers, offline: not teacher data. D2 rule p = .69. Code check vs rule arm rerun on the same topics (30/32): p = .18.

Control: two teachers accepted every rule, yet one had no say in how observations became rules. Since the pilot, teachers reword and rank rules and can see and switch off the recent-work summary.

Enforcement: a prompt did not enforce a standard checked across each set of options: the key-length rule did nothing as text (31/84 vs 32/84). Code caught it (17/84, p = .012) at 2.65x the calls; in the loop it halved the teacher's edits but still left 25/32 drafts to fix.

Judgement: the judge over-flags (0.89/0.61), the safer error in assessment: a valid item goes to review, not a flawed one to students. Code, reader and teacher each do what the others cannot.

https://fyp.z-chen.dev